MTH 201 Chapter 3: Infinite Series

3.1.1 Sequences

General Idea & Terminology:
We can think of a sequence as *an ordered list* of numbers. For example:

10,	20,	30,	40,	50,	…
1st thing $(n=1)$	2nd thing $(n=2)$	3rd thing $(n=3)$	4th thing $(n=4)$	5th thing $(n=5)$	Indicates that the list continues

NOTATION!
When talking about sequences (big lists of numbers) we usually write the following:
The sequence $\{a_1, a_2, a_3, \ldots\}$ can be denoted by: $a_n = f(n)$ where $f(n)$ is some function or $\{a_n\}_{n=1}^{\infty}$.
In general we will use a_n to represent the *general term* of a sequence by using a formula (like with our function above.

Note: A mathematical way to define an infinite sequence would be: An infinite sequence is a function whose domain is the natural numbers.

Example 1:
(a) Write the first 5 terms of the sequence given by $a_n = \dfrac{n}{n+1}$.

From the explicit formula we see that $a_1 = \dfrac{1}{2}$, $a_2 = \dfrac{2}{3}$, $a_3 = \dfrac{3}{4}$, $a_4 = \dfrac{4}{5}$, and $a_5 = \dfrac{5}{6}$.

(b) Find the terms: a_3, a_4, and a_5 of the sequence given by $a_n = \sqrt{n-3}$ where $n \geq 3$
From the explicit formula we see that $a_3 = 0$, $a_4 = 1$, and $a_5 = \sqrt{2}$.
Note: There is clearly some ambiguity here on what we should call the "first" term. We will ignore this for now.

(c) Find the first five terms of the following sequence which is given *recursively* as:
$f_1 = 1 \quad f_2 = 1 \quad f_n = f_{n-1} + f_{n-2}$ where $n \geq 3$. This sequence is known as the *Fibonacci sequence*.

First, we know $f_1 = 1$ and $f_2 = 1$. Then, by the recursive formula we have: $f_3 = f_2 + f_1 = 1 + 1 = 2$, $f_4 = 2 + 1 = 3$, and $f_5 = 3 + 2 = 5$.

Note: One big disadvantage to recursively defined sequences is that one must find certain previous terms in order to find more terms. For example, in the above example, one must find the 99th and 100th term before finding the 101st term.
Note: There is an awesome explicit formula for the Fibonacci sequence (usually proven in discrete math or combinatorics). It is: $f_n = \dfrac{1}{\sqrt{5}}\left[\left(\dfrac{1+\sqrt{5}}{2}\right)^n - \left(\dfrac{1-\sqrt{5}}{2}\right)^n\right]$.

Example 2 (you try): Find an explicit formula for the sequence whose first few terms are:

(a) 1, 8, 27, 64, 125, …

One possibility is $a_n = n^3$.

(b) 1, 3, 7, 15, 31, 63, 127, …

One possibility is $a_n = 2^n - 1$.

(c) $1, \frac{-1}{2}, \frac{1}{3}, \frac{-1}{4}, \frac{1}{5}, \frac{-1}{6}, \ldots$

One possibility is $a_n = \frac{(-1)^{n+1}}{n}$.

Note: In each part of this example one can easily check whether or not his/her solution works by finding the first few terms of the sequence that his/her explicit formula defines.

Question to the Class: If $a_n = \frac{n}{n+1}$, what do you think $\lim_{n\to\infty} a_n$ should equal?

Answer: One possible answer is $\lim_{n\to\infty} a_n = 1$ since the terms of the sequence get arbitrarily close to 1.

Note: If time permits we will graph this sequence on the calculator.

Definition: A sequence $\{a_n\}$ has the <u>limit</u> L and we write $\lim_{n\to\infty} a_n = L$ or $a_n \to L$ as $n \to \infty$ if for every $\varepsilon > 0$ there is a corresponding integer N such that if $n > N$ then $|a_n - L| < \varepsilon$.

Note: If L exists as a finite number we say that the sequence is convergent. Otherwise, the sequence is said to be divergent.

Note: Intuitively, this definition is saying that a sequence converges to L if we can always answer the question: "How far out must one go to guarantee that the remaining terms are within epsilon of L?"

Example 3: Use the above definition to prove that if $a_n = \frac{n}{n+1}$, then $\lim_{n\to\infty} a_n = 1$.

We begin with some scratch work and trying to understand the problem intuitively.

Let's begin by asking: How far out in the sequence must we go to guarantee that the remaining terms are within $\frac{1}{1000}$ of 1. To answer this we solve:

$\left|\frac{n}{n+1}-1\right| < \frac{1}{1000} \Leftrightarrow \left|\frac{-1}{n+1}\right| < \frac{1}{1000} \Leftrightarrow \frac{1}{n+1} < \frac{1}{1000} \Leftrightarrow 1000 < n+1 \Leftrightarrow 999 < n$. This shows that if we go beyond the 999$^{\text{th}}$ term in the sequence, any term will be within a thousandth of one.

Now, we ask the more general question: How far out in the sequence must we go to guarantee that the remaining terms are within ε of 1 (where $\varepsilon > 0$). To answer this we solve:

$\left|\frac{n}{n+1}-1\right| < \varepsilon \Leftrightarrow \left|\frac{-1}{n+1}\right| < \varepsilon \Leftrightarrow \frac{1}{n+1} < \varepsilon \Leftrightarrow 1 < \varepsilon(n+1) \Leftrightarrow \frac{1}{\varepsilon} - 1 < n$. This shows that if we go beyond $\left\lceil \frac{1}{\varepsilon} - 1 \right\rceil$ terms, any term will be within ε of 1.

Note: $\lceil x \rceil$ denotes the smallest integer greater than or equal to x.

Now, we are prepared to begin the proof.

Proof: Suppose ε is an arbitrary positive number and $N = \left\lceil \frac{1}{\varepsilon} - 1 \right\rceil$. We see that if $n > N$, then

$|a_n - 1| = \left|\frac{n}{n+1} - 1\right| = \left|\frac{-1}{n+1}\right| = \frac{1}{n+1} < \frac{1}{N+1} = \frac{1}{\left\lceil \frac{1}{\varepsilon} - 1 \right\rceil + 1} \le \frac{1}{\frac{1}{\varepsilon} - 1 + 1} = \frac{1}{\frac{1}{\varepsilon}} = \varepsilon$. Thus, we have that $\lim_{n\to\infty} a_n = 1$ by definition. ■

Definition: $\lim_{n\to\infty} a_n = \infty$ means that for every positive number M there is an integer N such that if $n > N$ then $a_n > M$.

Note: $\lim_{n\to\infty} a_n = -\infty$ can be defined similarly.

Theorem: If $\lim_{x\to\infty} f(x) = L$ and $f(n) = a_n$ when n is a positive integer, then $\lim_{n\to\infty} a_n = L$.

Proof: The proof follows from the definition of limit of a function.

Example 4:

Use the above theorem to show that if $a_n = \frac{n}{n+1}$, then $\lim_{n\to\infty} a_n = 1$.

Let $f(x) = \frac{x}{x+1}$. By l'Hospital's rule we know $\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left(\frac{x}{x+1}\right) = \lim_{x\to\infty}\left(\frac{1}{1}\right) = 1$. Thus, by the theorem $\lim_{n\to\infty} a_n = 1$.

■

Example 5 (you try):

Find with full justification: $\lim_{n\to\infty} \frac{\ln(n)}{n}$.

Let $f(x) = \frac{\ln(x)}{x}$. By l'Hospital's rule we know $\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left(\frac{\ln(x)}{x}\right) = \lim_{x\to\infty}\left(\frac{1/x}{1}\right) = 0$. Thus, by the theorem $\lim_{n\to\infty} \frac{\ln(n)}{n} = 0$.

■

The next theorem tells us that analogues of the limit laws we learned in Calculus I hold for sequences. They can be proven similarly to the limit laws presented in Calculus I.

Theorem: If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and c is a constant, then

1. $\lim_{n\to\infty}(a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$
2. $\lim_{n\to\infty} ca_n = c\lim_{n\to\infty} a_n$
3. $\lim_{n\to\infty}(a_n \cdot b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$
4. $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$ if $b_n \neq 0$
5. $\lim_{n\to\infty} a_n^p = \left[\lim_{n\to\infty} a_n\right]^p$ if $p > 0$ and $a_n > 0$.

Theorem: If $a_n \le b_n \le c_n$ for $n \ge n_0$ and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$.

Note: This is the analogue of the squeeze theorem and its proof is similar to the proof of the squeeze theorem.

Theorem: If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Proof (not done in class): If $\lim_{n\to\infty} |a_n| = 0$, we know by definition that for all $\varepsilon > 0$ there is a positive integer N such that if $n > N$ then $\left||a_n| - 0\right| < \varepsilon$. So, for an arbitrary positive number ε' we know there is a positive integer N' such that if $n > N'$ then $\left||a_n| - 0\right| < \varepsilon'$.

From these facts we note that if $n > N'$, then $|a_n - 0| = \left||a_n|\right| = \left||a_n| - 0\right| < \varepsilon'$. Since ε' was arbitrary, we now know that for all $\varepsilon > 0$ there is a positive integer N such that if $n > N$ then $|a_n - 0| < \varepsilon$. Thus, $\lim_{n\to\infty} a_n = 0$ by definition.

■

Note: This theorem can be useful when a sequence has terms that alternate between negative and positive.

Example 6:

Evaluate $\lim_{n\to\infty}\left(\frac{(-1)^n}{n}\right)$ if it exists.

We first note that $\left|\frac{(-1)^n}{n}\right| = \frac{1}{n}$. We know from chapter 2 that if $f(x) = \frac{1}{x}$, then $\lim_{x\to\infty} f(x) = 0$.

Thus, we know that $\lim_{n\to\infty}\frac{1}{n} = 0$, and by the above theorem $\lim_{n\to\infty}\left(\frac{(-1)^n}{n}\right) = 0$.

Example 7:

Determine whether each sequence converges or diverges (we implicitly assume that this means as *n* goes to infinity). If it converges find the exact limit. If it diverges provide an explanation as to why it diverges.

(a) $a_n = (-1)^n$

Since the terms of this sequence oscillate between -1 and 1 infinitely often, this sequence does not approach any number. Thus, $\lim_{n\to\infty} a_n$ does not exist, and the sequence is divergent.

(b) $b_n = \sqrt{\dfrac{n+1}{9n+1}}$

If $f(x) = \sqrt{\dfrac{x+1}{9x+1}}$, then by the properties of limits and l'Hopital's rule we have $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} \sqrt{\dfrac{x+1}{9x+1}} = \sqrt{\lim\limits_{x\to\infty} \dfrac{x+1}{9x+1}} = \sqrt{\lim\limits_{x\to\infty} \dfrac{1}{9}} = \sqrt{\dfrac{1}{9}} = \dfrac{1}{3}$. So, by the first theorem presented in this lecture $\lim\limits_{n\to\infty} b_n = \dfrac{1}{3}$, and the sequence is convergent.

(c) $c_n = \dfrac{(-1)^{n+1} n}{n+\sqrt{n}}$

We first note that $|c_n| = \left|\dfrac{(-1)^{n+1} n}{n+\sqrt{n}}\right| = \dfrac{n}{n+\sqrt{n}}$. Now, if $f(x) = \dfrac{x}{x+\sqrt{x}}$, we know by l'Hopital's rule that $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} \dfrac{x}{x+\sqrt{x}} = \lim\limits_{x\to\infty} \dfrac{1}{1+\dfrac{1}{2\sqrt{x}}} = \lim\limits_{x\to\infty} \dfrac{2\sqrt{x}}{2\sqrt{x}+1} = \lim\limits_{x\to\infty}\left(1-\dfrac{1}{2\sqrt{x}+1}\right) = 1$. Thus we have $\lim\limits_{n\to\infty} |b_n| = 1$, and the last theorem we learned in this class does not apply. Since c_n alternates between positive and negative terms (the even terms are negative and the odd terms are positive) and $\lim\limits_{n\to\infty} |c_n| = 1$, we know that the odd terms of c_n approach 1 and the even terms of c_n approach -1. Thus, $\lim\limits_{n\to\infty} c_n$ does not exist, and the sequence is divergent.

Theorem: If $\lim_{n\to\infty} a_n = L$ and the function f is continuous at L, then

$$\lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right) = f(L) \ .$$

Proof: This fact follows from the epsilon delta definition of continuity (details will be omitted).

Example 8: Determine whether $a_n = \cos\left(\frac{2}{n}\right)$ is convergent or divergent.

Since $\lim_{n\to\infty} \frac{2}{n} = 0$ and the cosine function is continuous, we have by the above theorem:
$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \cos\left(\frac{2}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{2}{n}\right) = \cos(0) = 1$. Thus, the sequence is convergent.

Theorem: The sequence $\{r^n\}$ is convergent if $-1 < r \le 1$ and divergent for all other values of r.
Moreover, $\lim_{n\to\infty} r^n = \begin{cases} 0 & if \quad -1 < r < 1 \\ 1 & if \quad r = 1 \end{cases}$.

Example 9: Determine (with justification) whether $a_n = \frac{9^{n-1}}{10^{2n+1}}$ converges or diverges.

By the laws of limits for sequences, we see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{9^{n-1}}{10^{2n+1}}\right) = \lim_{n\to\infty} \left(\frac{9^n \cdot 9^{-1}}{100^n \cdot 10}\right) = \frac{9^{-1}}{10} \lim_{n\to\infty} \left(\left[\frac{9}{100}\right]^n\right)$.

Now, by the most recent theorem we know $\lim_{n\to\infty} \left(\left[\frac{9}{100}\right]^n\right) = 0$. Thus, $\lim_{n\to\infty} a_n = 0$, and the sequence is convergent.